

**Queue Project AC1-126, AC1-127, AC1-128, AC1-129, AC1-131,
AC1-132, and AC1-133
System Impact Study Report**

Network Impacts

The following projects were evaluated.

QueueNumber	OasisNumber	CustomerName	PathName	MW
AC1-126	4841985	North Carolina Electric Membership	PJM-CPLE	25
AC1-127	4841988	North Carolina Electric Membership	PJM-CPLE	25
AC1-128	4841989	North Carolina Electric Membership	PJM-CPLE	25
AC1-129	4841990	North Carolina Electric Membership	PJM-CPLE	25
AC1-131	4841992	North Carolina Electric Membership	PJM-CPLE	50
AC1-132	4841993	North Carolina Electric Membership	PJM-CPLE	50
AC1-133	4841994	North Carolina Electric Membership	PJM-CPLE	100

Summer Peak Analysis - 2020

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

1. (DVP - CPLE) The 3BTLEBRO-3ROCKYMT115T 115 kV line (from bus 314554 to bus 304223 ckt 1) loads from 151.52% to 156.84% (AC power flow) of its emergency rating (164 MVA) for the tower line contingency outage of 'LN 2058-2181'. This project contributes approximately 8.97 MW to the thermal violation.

CONTINGENCY 'LN 2058-2181'

```
OPEN BUS 304226                /* ISLAND: 6PA-RMOUNT#4115.00
OPEN BRANCH FROM BUS 304226 TO BUS 314591 CKT 1      /* 6PA-RMOUNT#4230.00 - 6NASH 230.00
OPEN BRANCH FROM BUS 313845 TO BUS 314591 CKT 1      /* 6HATHAWAY 230.00 - 6NASH 230.00
OPEN BUS 314591                /* ISLAND: 6NASH 230.00
OPEN BRANCH FROM BUS 304222 TO BUS 313845 CKT 1      /* 6ROCKYMT230T230.00 - 6HATHAWAY 230.00
```

END

Please refer to Appendix 1 for a table containing the generators having contribution to this flowgate.

Affected System Analysis & Mitigation

External PJM Impacts:

External PJM Impacts to be determined during later study phases (as applicable).

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

Not Applicable

Light Load Analysis - 2020

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

1. (PJM500 - PJM500) The PEACHBTM-CNASTONE 500 kV line (from bus 200013 to bus 200004 ckt 1) loads to 118.43% (DC power flow) of its normal rating (2338 MVA) for non-contingency condition. This project contributes approximately 40.97 MW to the thermal violation.

Please refer to Appendix 2 for a table containing the generators having contribution to this flowgate.

System Reinforcements

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

1. (DVP - CPLE) The 3BTLEBRO-3ROCKYMT115T 115 kV line (from bus 314554 to bus 304223 ckt 1) loads from 151.52% to 156.84% (AC power flow) of its emergency rating (164 MVA) for the tower line contingency outage of 'LN 2058-2181'. This project contributes approximately 8.97 MW to the thermal violation.

In order to mitigate the above violation, the Battleboro to Rocky Mount circuit needs to be reconductored to 1590 ACSR and replace terminal equipment at Rocky Mount substation needs to be replaced.

Reconductoring the circuit to 1590 ACSR and upgrading the terminal equipment at Rocky Mount (\$15 Million):

Queue	MW contribution	Percentage of Cost	Cost(\$15.00M)	Contingency Name	Contingency Type
AB2-059	24.577	21.90%	\$3.28	LN 2058-2181'	tower

AB2-100	7.96	7.09%	\$1.06	LN 2058-2181'	tower
AB2-128	6.37	5.68%	\$0.85	LN 2058-2181'	tower
AB2-171	5.12	4.56%	\$0.68	LN 2058-2181'	tower
AB2-174	6.80	6.06%	\$0.91	LN 2058-2181'	tower
AC1-034	13.41	11.95%	\$1.79	LN 2058-2181'	tower
AC1-086	21.59	19.23%	\$2.89	LN 2058-2181'	tower
AC1-099	17.45	15.54%	\$2.33	LN 2058-2181'	tower
AC1-133	8.97	7.99%	\$1.20	LN 2058-2181'	tower

Though this project is not responsible for replacing the breaking leads at Battleboro Substation, this project depends on the upgrades from AB2 queue projects i.e. replacing the breaker leads at Battleboro substation.

Light Load Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

1. To relieve the Peach Bottom to Conastone 500 kV line overload

Existing baseline B2766.1; Upgrade substation equipment at Peach Bottom 500 kV (on the Peach Bottom – Conastone 500 kV circuit) to increase facility rating to 2826 MVA normal and 3525 MVA emergency. AC1-133 will require B2766.1 to be advanced from 2021 to 2020. AC1-133 is responsible for the cost required to advance B2766.2 from 2021 to 2020.

Cost: PECO advancement cost = \$0.

Existing baseline B2766.2; Upgrade substation equipment at Conastone 500 kV (on the Peach Bottom – Conastone 500 kV circuit) to increase facility rating to 2826 MVA normal and 3525 MVA emergency. AC1-133 will require B2766.2 to be advanced from 2021 to 2020. AC1-133 is responsible for the cost required to advance B2766.2 from 2021 to 2020.

Cost: There is no cost for advancing B2766.2 from 2021 to 2020 at this point. Further evaluation will need to be performed during the Facility Study Phase in order to determine whether BGE requires additional cost for the advancement of this upgrade.

Appendices

The following appendices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gage other generators impact.

It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

Appendix 1

(DVP - CPLE) The 3BTLEBRO-3ROCKYMT115T 115 kV line (from bus 314554 to bus 304223 ckt 1) loads from 151.52% to 156.84% (AC power flow) of its emergency rating (164 MVA) for the tower line contingency outage of 'LN 2058-2181'. This project contributes approximately 8.97 MW to the thermal violation.

CONTINGENCY 'LN 2058-2181'

```

OPEN BUS 304226                /* ISLAND: 6PA-RMOUNT#4115.00
OPEN BRANCH FROM BUS 304226 TO BUS 314591 CKT 1      /* 6PA-RMOUNT#4230.00 - 6NASH 230.00
OPEN BRANCH FROM BUS 313845 TO BUS 314591 CKT 1      /* 6HATHAWAY 230.00 - 6NASH 230.00
OPEN BUS 314591                /* ISLAND: 6NASH 230.00
OPEN BRANCH FROM BUS 304222 TO BUS 313845 CKT 1      /* 6ROCKYMT230T230.00 - 6HATHAWAY 230.00
END

```

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
315131	1EDGECEMA	2.59
315132	1EDGECEMB	2.59
315139	1GASTONA	2.51
315141	1GASTONB	2.51
315126	1ROARAP2	1.08
315128	1ROARAP4	1.04
315134	1ROAVALA	3.56
315135	1ROAVALB	0.95
315136	1ROSEMG1	2.02
315138	1ROSEMG2	0.95
315137	1ROSEMS1	1.25
900671	V4-068 C	0.07
900672	V4-068 E	0.19
917331	Z2-043 C	0.38
917332	Z2-043 E	0.86
917341	Z2-044 C	0.56
917342	Z2-044 E	1.26
917511	Z2-088 C OP1	0.73
917512	Z2-088 E OP1	6.22
917591	Z2-099 C	0.12
917592	Z2-099 E	0.28
918411	AA1-050	0.61
LTF	AA1-055	9.32
921162	AA1-063AC	5.73
921163	AA1-063AE	2.7
918512	AA1-065 E OP	2.03
921183	AA1-067 E	0.32
918561	AA1-072 C	0.06

918562	AA1-072 E	0.14
921752	AA2-053 C	5.82
921753	AA2-053 E	2.5
921762	AA2-057 C	13.08
921763	AA2-057 E	6.54
921862	AA2-068 C	3.37
921863	AA2-068 E	1.55
920021	AA2-086 C	0.06
920022	AA2-086 E	0.15
921982	AA2-088 C	4.15
921983	AA2-088 E	6.78
922442	AA2-165 C	1.78
922443	AA2-165 E	0.86
922472	AA2-169 C	0.92
922473	AA2-169 E	0.42
922512	AA2-174 C	0.27
922513	AA2-174 E	0.29
922922	AB1-081 C OP	20.25
922923	AB1-081 E OP	8.68
923262	AB1-132 C OP	10.08
923263	AB1-132 E OP	4.32
923572	AB1-173 C OP	1.29
923573	AB1-173 E OP	0.6
923582	AB1-173AC OP	1.29
923583	AB1-173AE OP	0.6
923911	AB2-031 C OP	1.28
923912	AB2-031 E OP	0.63
923941	AB2-035 C	0.38
923942	AB2-035 E	0.16
923991	AB2-040 C OP	4.2
923992	AB2-040 E OP	3.44
924151	AB2-059 C	23.86
924152	AB2-059 E	12.29
924381	AB2-087 C	0.34
924382	AB2-087 E	0.16
924391	AB2-088 C	0.48
924392	AB2-088 E	0.23
924491	AB2-098 C	0.25
924492	AB2-098 E	0.11
924501	AB2-099 C	0.35
924502	AB2-099 E	0.15
924511	AB2-100 C	5.52
924512	AB2-100 E	2.72
925122	AB2-169 E	2.31
925171	AB2-174 C OP	3.78

925172	<i>AB2-174 E OP</i>	3.42
926331	<i>AC1-034 C OP</i>	7.64
926332	<i>AC1-034 E OP</i>	5.76
926851	<i>AC1-086 C</i>	14.84
926852	<i>AC1-086 E</i>	6.75
926981	<i>AC1-099 C</i>	10.95
926982	<i>AC1-099 E</i>	6.5
<i>LTF</i>	<i>AC1-133</i>	8.97
927731	<i>AC1-163 C</i>	1.27
927732	<i>AC1-163 E</i>	0.44
927991	<i>AC1-189 C</i>	5.28
927992	<i>AC1-189 E</i>	2.63
928171	<i>AC1-206 C OP</i>	4.45
928172	<i>AC1-206 E OP</i>	2.12
928191	<i>AC1-208 C O1</i>	12.83
928192	<i>AC1-208 E O1</i>	5.7

Appendix 2

(PJM500 - PJM500) The PEACHBTM-CNASTONE 500 kV line (from bus 200013 to bus 200004 ckt 1) loads to 118.43% (DC power flow) of its normal rating (2338 MVA) for non-contingency condition. This project contributes approximately 40.97 MW to the thermal violation.

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
204652	27PORT1GEN	19.68
204653	27PORT2GEN	28.75
206325	28O C GEN	189.25
219230	EAGLEPT_G1	3.19
292769	U1-010	2.54
292934	U2-015C OP1	2.34
292935	U2-015E OP1	15.67
292966	U2-045 C	0.68
292967	U2-045 E	4.57
292070	V1-026	1.97
292071	V1-027	2.03
900001	V4-019	0.86
901001	W1-108	1.28
902401	W2-028	1.59
902281	W2-029	1.64
903201	W3-032A 1	17.47
903211	W3-032A 2	17.47
905531	W4-063 C	0.17
905532	W4-063 E	1.14
907011	X1-068	0.9
907991	X1-078	1.47
907323	X1-096 C	0.78
907324	X1-096 E	18.94
912042	X4-005 E	17.33
218403	X4-044 E	2.22
913101	Y1-026	13.87
913341	Y1-077	10.82
915191	Y3-043	109.5
916131	Z1-057	1.76
916141	Z1-058	3.12
916151	Z1-059 C	1.56
916152	Z1-059 E	1.4
916431	Z1-099	2.55
917011	Z2-002 C	4.86
917012	Z2-002 E	4.2

917521	Z2-089 C	44.42
917522	Z2-089 E	16.6
918191	AA1-034 C	9.87
918192	AA1-034 E	22.73
918571	AA1-073 C	1.1
918572	AA1-073 E	0.48
919231	AA1-079_1	11.71
919230	AA1-079_2	11.71
919531	AA2-017 C	2.4
919532	AA2-017 E	16.11
919681	AA2-052	0.26
919802	AA2-066 E	0.59
919831	AA2-069	193.25
919931	AA2-077	69.83
920301	AA2-128 C OP	20.36
920302	AA2-128 E OP	20.36
920421	AA2-140	11.17
919871	AB1-000 1	1.96
919872	AB1-000 2	1.96
919873	AB1-000 3	1.96
930201	AB1-056 C OP	18.72
930202	AB1-056 E OP	53.3
930211	AB1-057 C	19.01
930212	AB1-057 E	54.17
931051	AB1-154 C	257.57
931052	AB1-154 E	16.09
923791	AB2-014	2.45
924111	AB2-055 OP	289.38
924341	AB2-082 C	189.12
924342	AB2-082 E	1.13
924421	AB2-091 C	27.62
924422	AB2-091 E	1.38
924431	AB2-092	14.12
924852	AB2-139 E	0.13
925181	AB2-175	10.
925182	AB2-175 C2	10.
926041	AC1-029	5.8
926141	AC1-066 C	0.56
926142	AC1-066 E	0.84
926161	AC1-071 C	1.54
926162	AC1-071 E	10.33
926561	AC1-184	11.17